REPORT 010: NKEDITOR4 — EPOCH 010: INTERACTIVE 8OPPERC PERFORMANCE COCKPIT VIEW IN RUST EGUI

2026-11-04 01:02:31 | Interactive 8opPerc Performance Cockpit View (src/op8_cockpit.rs) with 8-Column Percussion Voice Matrix, Timbre Lock, and Bidirectional OSC/LED Wiring

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Prompt

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Build the interactive 8opPerc Performance Cockpit view in the Rust egui application
↪  (`src/op8_cockpit.rs`): render an 8-column Percussion Voice Matrix aligned with
↪  the 8 physical tracks of the Korg nanoKONTROL2. For each voice lane, display its
↪  family name, active timbre patch badge, queued next-bar timbre indicator,
↪  interactive density ladder meter (`0.1x` to `5.0x`), level/decay macro slider,
↪  Solo/Mute/Rec LED buttons, timbre Lock toggle (preventing global timbre set
↪  changes from overwriting locked lanes), and one-click Timbre Reroll button. Wire
↪  all cockpit interactions bidirectionally to the OSC server and hardware LED
↪  controller so GUI clicks update both the Konduktiva rig and the physical
↪  nanoKONTROL2 LEDs. Add automated tests (with `CARGO_BUILD_JOBS=2`), git commit
↪  all changes, and compile `reports/report-010.html` from `reports/report-010.md`
↪  via `/home/renick/claude/html-report-generator/build-report.sh --pdf` and commit.
```

TL;DR

Implemented the interactive 8opPerc Performance Cockpit view in Rust egui (src/op8_cockpit.rs), rendering an 8-column Percussion Voice Matrix aligned with the 8 physical tracks of the Korg nanoKONTROL2. Each lane displays its family name, active and queued next-bar timbre badges, interactive density ladder meter (0.1x to 5.0x), level/decay macro slider, Solo/Mute/Rec LED buttons, Timbre Lock toggle, and one-click Timbre Reroll button, wired bidirectionally to OscServer and MultiDeviceLedManager. All 41 Rust tests and 10 Node.js tests pass (51 total).

Plain-English Overview: What Changed and Why It Matters

When performing live with the `8opPerc` percussion rig and a Korg `nanoKONTROL2`, the performer needs an at-a-glance visual dashboard ("cockpit") on their computer screen that lines up column-for-column with the 8 physical fader strips under their fingers. Without a visual cockpit, the performer cannot see which sound patch (`timbre`) is currently loaded on each drum voice, which new sound is waiting to drop on the next bar downbeat (`queued_timbre`), or what the exact note density multiplier (`0.1x` to `5.0x`) is for each lane.

In Epoch 10, we built `src/op8_cockpit.rs` to provide this interactive `egui` Performance Cockpit:

- **8-Column Percussion Voice Matrix Aligned with Korg nanoKONTROL2**: Renders 8 vertical voice columns (`T1 KICK` through `T8 CYMBAL`) styled with `yaae-theme` and `twoBowls` crisp zero-rounding borders (`Rounding::ZERO`). Each column shows the voice's family name, its Konduktiva live role (`kicks`, `snares`, `hats`, `toms`, `percussion`, `alt-toms`, `noise`, `fx`), and an `AUDIBLE` vs `SILENCED` status pill that reflects both Mute and Solo priority.
- **Active & Next-Bar Queued Timbre Badges**: Displays the currently sounding FM percussion patch alongside the queued patch scheduled for the next barline.

- **Interactive Density Ladder Meter & Macro Slider**: Combines a custom vector progress bar and logarithmic slider (`0.10x` to `5.00x`, displaying both per-lane density and effective `global * lane` density) with a `0..127` level/decay macro slider.
- **Per-Lane Timbre Lock Toggle**: Added `timbre_locked` to `Op8VoiceLane` and a `LOCKED / UNLOCKED` button on every column. When a performer locks a favorite kick or snare sound, pressing `Cycle` (`CC 46`) to switch the full 8-voice timbre set updates all unlocked voices while preserving the locked voices untouched.
- **Bidirectional OSC & Hardware LED Synchronization**: Clicking any Solo (`S`), Mute (`M`), Rec/Reroll (`R`), Density, Form, or Snapshot control in the `egui` cockpit immediately updates the physical LEDs on the Korg `nanoKONTROL2` via `MultiDeviceLedManager` and broadcasts the corresponding Open Sound Control (`OSC`) message to `Konduktiva` via `OscServer`.

Plain-English Cockpit Element	Code Component (‘src/op8_cockpit.rs‘)	Hardware Control / LED	OSC Broadcast Endpoint
Cockpit State & Controller	`Op8CockpitViewState`	Syncs all 28 `nanoKONTROL2` LEDs	`/nkeditor/op8/state`
8-Column Matrix Renderer	`render_op8_cockpit` / `render_voice_lane_column`	Aligned 1:1 with Tracks `1..8`	All `/nkeditor/op8/*` endpoints
Density Ladder Meter (`0.1x..5.0x`)	`Op8CockpitViewState::set_lane_density`	Track Knobs `CC 16..23`	`/nkeditor/op8/lane/ /density`
Level / Decay Macro Slider	`Op8CockpitViewState::set_lane_gain_cc`	Track Sliders `CC 0..7`	`/nkeditor/control/ set`
Solo Button (`S`)	`Op8CockpitViewState::toggle_lane_solo`	Solo Buttons & LEDs `CC 32..39`	`/nkeditor/op8/lane/ /solo`
Mute Button (`M`)	`Op8CockpitViewState::toggle_lane_mute`	Mute Buttons & LEDs `CC 48..55`	`/nkeditor/op8/lane/ /mute`
Single-Lane Timbre Reroll (`R`)	`Op8CockpitViewState::reroll_lane_timbre`	Rec Buttons & Flash LEDs `CC 64..71`	`/nkeditor/op8/ timbre`
Timbre Lock Toggle	`Op8CockpitViewState::toggle_lane_timbre_lock`	Protects lane during `Cycle CC 46`	Preserved in `/nkeditor/op8/state`

Tests

QA

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**Headless `egui` Verification**: Verified `render_op8_cockpit` across a `1440x900` headless `egui::Context` frame with `ThemeSettings::default()`, confirming vector shape emission for all 8 voice columns and header controls without opening a window.

**UDP Loopback OSC & Hardware LED Verification**: Confirmed that toggling Solo/Mute, rerolling timbres, and adjusting lane density in `Op8CockpitViewState` simultaneously updates `DeviceLedController` (`CC 33`, `CC 50`, `CC 64` LEDs) and emits UDP OSC datagrams (`/nkeditor/op8/lane/1/solo`, `/nkeditor/op8/lane/2/mute`, `/nkeditor/op8/timbre`, `/nkeditor/op8/lane/0/density`, `/nkeditor/op8/state`).
```

Files touched

- ‘src/op8_cockpit.rs‘ (Created — 852 lines: ‘Op8CockpitViewState‘, ‘render_op8_cockpit‘, ‘render_cockpit_header_bar‘, ‘render_voice_lane_column‘, and 3 unit/integration tests)
- ‘src/op8_model.rs‘ (Modified — added ‘timbre_locked: bool‘ to ‘Op8VoiceLane‘ and updated ‘advance_full_timbre_set‘ to skip locked lanes)
- ‘src/lib.rs‘ (Modified — exported ‘pub mod op8_cockpit;‘)

Test coverage (51 total)

QA verdict

PASS

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